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MILITARY SPECIFICATION

# TOOL OUTFIT, HYDRAULIC SYSTEMS TEST AND REPAIR, MOBILE

This specification is approved for use by the US Army Armament Research, Development and Engineering Center, Department of the Army, and is available for use by all Department and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a trailer mounted hydraulic system test and repair tool outfit.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| QQ-S-781 | - | Strapping, Steel, and Seals |
| PPP-B-636 | - | Boxes, Shipping, Fiberboard |
| PPP-P-40 | - | Preservation and Packing of Hand Tools; Tools and Tool Accessories For Power Driven, Metal and Woodworking Machinery |

MILITARY

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4940
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

| | | |
|---|---|---|
| MIL-P-116 | - | Preservation, Methods of |
| MIL-B-121 | - | Barrier Material, Greaseproofed, Waterproofed, Flexible |
| MIL-L-2104 | - | Lubricating Oil, Internal Combustion Engine, Tactical Service |
| MIL-T-22085 | - | Tapes, Pressure-sensitive, Adhesive, Preservation and Sealing |
| MIL-L-46167 | - | Lubricating Oil, Internal Combustion Engine, Arctic |
| MIL-V-62038 | - | Vehicles, Wheeled: Preparation For Shipment and Storage of |

STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

| | | |
|---|---|---|
| MIL-STD-129 | - | Marking for Shipment and Storage |
| MIL-STD-130 | - | Identification Marking of US Military Property |
| MIL-STD-171 | - | Finishing of Metal and Wood Surfaces |
| MIL-STD-209 | - | Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment |
| MIL-STD-642 | - | Identification Marking of Combat and Tactical Transport Vehicles |
| MIL-STD-810 | - | Environmental Test Methods and Engineering Guidelines |
| MIL-STD-889 | - | Dissimilar Metals |
| MIL-STD-1190 | - | Minimum Guidelines for Level C Preservation, Packing and Marking |
| MIL-STD-1474 | - | Noise Limits for Army Materiel |

DRAWINGS

AL

| | |
|---|---|
| 13221E6705 | - Door Lock, Recessed Handle |
| 13221E6850 | - Tool Outfit, Hydraulic System Test and Repair, Mobile. |
| 13226E7200 | - Hydraulic Systems Test and Repair Unit. |

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US DEPARTMENT OF LABOR

| | | |
|---|---|---|
| 29 Code of Federal Regulation (CFR), Chapter XVII, Parts 1900 to 1910 | - | Occupational Safety and Health Standards Act (OSHA) |

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as

directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

ASSOCIATION OF AMERICAN RAILROADS

Rules Governing the loading of Department of Defense Materiel on Open Top Car.

(Application for copies should be addressed to the American Railroad Association, 1920 L. Street, NW, Washington, DC 20036.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P. Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The tool outfit shall be in accordance with Top Assembly 13221E6850 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the tool outfit. Any data, e.g., shop drawings, layouts, flow sheets, processing procedures, etc., prepared by the contractor or from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1). The approved first article and the production item shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of the specification.

3.3 Initial production. When specified (see 6.2.1), the contractor shall furnish to the Government one or more tool outfits for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specifically designated herein or in the contract shall be of a quality commensurate with best commercial practices, shall be suitable for the intended purpose in the design and fabrication of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would affect the performance, maintainability, reliability, durability, or longevity of the individual components or the end item equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.5 Chassis. The Government-furnished M116A1 3/4 ton, 2-wheel, trailer chassis shall be modified by the contractor in accordance with Top Assembly 13221E6850 and as specified herein.

3.6 Body. The contractor shall fabricate and install the body on the chassis in accordance with the applicable drawings. The contractor shall furnish all items of equipment specified on the drawings.

3.7 Electrical interface. All electrically actuated items shall operate without malfunction when the tool outfit is connected to 110/120 volts 50-60 cycles per second (Hz) alternating current power from military standard generators and commercial power sources.

3.8 Sliding tables, doors, and support legs. The sliding tables, doors, and support legs shall function smoothly, without binding and there shall be no evidence of cutting or pinching of hose or electrical wires or cables in sliding mechanisms or hinge points.

3.9 Wash rack. The wash rack tray shall operate smoothly without binding and without cutting or pinching of hose or electrical wires or cables. The solvent pumping equipment shall operate without evidence of leakage of malfunction.

3.10 Proof pressure tester. The proof pressure tester shall supply oil at a pressure of 70,000 kilopascal (kPa) plus or minus 3500 kPa without evidence of leakage or malfunction.

3.11 Transportability.

3.11.1 Lifting and tie-down provisions. The tool outfit, shall conform to the sling lifting and tie-down requirements of MIL-STD-209 for Type II equipment, Class 1 and 2.

3.11.2 Railroad transportability. The tool outfit with all components properly stored for shipment shall withstand a Rail Impact Test IAW MIL-STD-810 without damage or permanent deformation, at impact velocities of 7, 10, and 13 kilometers per hour (km/h) in forward direction, and 13 km/h in the reverse direction.

3.12 Roadability. When loaded with the tools and equipment specified herein, the tool outfit shall withstand towing on paved roads, gravel roads, cross-country trails, Belgian block, 18.5 degree side slopes and 20 degree ramps, when tested as specified in 4.6.2.1. The towing shall include abrupt stops and sharp turns. During and following the towing, there shall be no damage to or displacement of (see 6.6.1) any component, accessory, part, or tool installed in or on the tool outfit; and there shall be no evidence of defective welded, bolted or riveted connections on the tool outfit.

3.13 Safety. All exposed parts which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed, or guarded. Guards and other appropriate protection shall be provided for the operator(s) of the tools in accordance with appropriate Occupation Safety and Health Regulations.

3.14 Environmental. The tool outfit equipment shall start and operate at temperatures ranging from 255 Kelvin (K) to 320 K. The tool outfit equipment shall not be damaged nor its operation impaired when stored at an ambient temperature of 340 K for 4 hours and at 230 K for a continuous period of 6 hours. The tool outfit and equipment shall show no evidence of damage, malfunction or irregular operation when tool outfit is subjected to extreme conditions of rain, wind, and humidity.

3.15 Rough handling. The tool outfit shall withstand stresses without damage to or displacement of any component when exposed to shock and vibration, such as occurring in rail transportation.

3.16 Tools and equipment. The tools and equipment shall be stored in tool boxes where size permits. The tools and equipment shall be anchored to prevent movement during towing and transit as specified herein. Tool anchorage devices shall be constructed for general tool or instrument configuration to allow interchange of like items from different manufacturing sources.

3.17 Locks. The contractor shall furnish key-locking device as specified in drawing 13221E6705. In lots of fifty, each tool outfit shall have different locks and keys from all other tool outfits in the same lot.

3.18 Noise limits. The noise limits of the tool outfit shall not exceed the limits specified in MIL-STD-1474, category C for steady state noise when tested in accordance with 4.6.2.8 (aural non-detectability and impulse noise do not apply). The precautions of MIL-STD-1474 shall be provided if the sound level exceeds 85 dB(A) when tested in accordance with 4.6.2.8. Hearing protectors shall not be supplied with the tool outfit. Hazard signs, when required, shall state "CAUTION, HEARING PROTECTION REQUIRED."

3.19 Identification marking and instruction plates.

3.19.1 Identification marking. The tool outfit shall be identified in accordance with MIL-STD-130, except color of marking shall conform to MIL-STD-642.

3.19.2 Vehicle marking. Registration and other marking shall be applied in accordance with MIL-STD-642.

3.20 Treatment and painting. The tool outfit body and all other items requiring painting internally and externally shall be cleaned, treated, and painted in accordance with finish 5.1.1 or 7.3.1 plus 24.17 (interior, ferrous or aluminum alloys) and 5.1.1 or 7.3.1 plus 20.24 (exterior, ferrous or aluminum alloys) of MIL-STD-171 as applicable. The paint application and testing shall be IAW MIL-C-53072. Unless otherwise specified, color shall be 34094 of FED-STD-595. The external areas of the body and trailer shall be painted in a camouflage pattern IAW drawing 13226E7200.

3.21 Government-furnished property. The following property in the quantities indicated will be furnished by the Government (see 6.4).

| Item No. | Description | National Stock No. | Quantity furnished each outfit tested |
|---|---|---|---|
| 1 | Chassis, Trailer, (military design) M116A2, 3/4-ton, 2-wheel | 2330-01-101-8434 | 1 |

3.22 Government-loaned property. The following property in the quantities indicated will be loaned by the Government for testing (see 6.5):

| Item No. | Description | National Stock No. | Quantity furnished each outfit tested |
|---|---|---|---|
| 1 | Truck, cargo, 1-1/4-ton, 4 x 4, (minimum size truck) | 2320-00-579-8942 | 1 |
| 2 | Generator set, 5 kW 60-cycle | 6115-00-017-8240 | 1 |

3.23 Endurance. The tool outfit shall perform twenty missions without failure when tested as specified in 4.6.2.3.

3.24 Workmanship. The quality of workmanship imparted to the item and all of its components shall equal or exceed that typically provided to commercial products by domestic producers of the type of item addressed herein. The items presented for acceptance shall have been manufactured with skill and care; shall be uniform, neat, and clean; and shall be free of irregularities and anomalies which degrade form, fit, function, performance or appearance.

3.24.1 Fabrication. Metal used in the fabrication of the tool outfit shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Flame cutting of steel may be employed instead of shearing or sawing. Burned surfaces of flame-cut steel shall be ground or machined to remove ash and cooling checks. All burrs, sharp edges and rough edges shall be removed. Overheating shall be avoided. All bends of a major character shall be made with metal dies or fixtures to ensure uniformity of size and shape.

3.25 Mercury restriction. The tool outfit shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.26 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the tool outfit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.2 Disassembly inspection. Failure of any test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with the specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected.

4.1.3 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products submitted to the government for acceptance comply with all the requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Initial production inspection (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Inspection comparison (see 4.7).
e. Packing inspection (see 4.8).

4.3 First Article inspection.

4.3.1 Examination. Examine the tool outfit as specified in 4.6.1. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

4.3.2 Tests. Test the tool outfit as specified in 4.6.1. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production tool outfits will be selected at random by the Government from the tool outfits being produced by production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2 to determine conformance to the

requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production tool outfit shall not exclude the remaining tool outfits from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.

washboarding and rutting and occasional potholes.

c. Cross-country trails. Tow the tool outfit 240 km on cross-country trails at maximum safe speeds. The roadbed shall consist of native soil with rock added as required to keep the road passable. Surface conditions shall range from smooth to rough with washboarding, rutting, potholes, and depressed areas.
d. Belgian block. Tow the tool outfit 80 km over Belgian block at 13 km/h.
e. Diagonal ditch course. Tow the tool outfit over a series of ditches until 600 kPa reversal torsion frame stresses have been applied. The ditch profile shall be such as to allow vertical wheel travel of 23 centimeter (cm) from crest to root. The ditches shall be at 45 degrees to the centerli of the ou se nd spac d i nt ist p o l w nly one wheel in the ditches at any one time.
f. Stops. Fifty normal stops shall be made during paved highway testing (a.) above. In addition, five panic stops from 32 km/h shall be made on a clean level pavement.
g. Turns. Fifty sharp right-hand and 50 sharp left-hand turns through 720 degrees per SAE J695.
h. Side slopes. The tool outfit will be towed in each direction of travel across slopes 11.4 degrees and 18.5 degrees.
i. Ramp. The tool outfit will be towed up and down a 20 degree ramp.

Examine the tool outfit at the completion of each phase of this test. Damage to or displacement of (see 6.6.1) any component, accessory, part or tool; or evidence of defective welded, bolted, riveted connections, shall constitute failure of this test.

4.6.2.2 Rail Impact. The Tool Outfit (see 3.11.2) shall be subjected to a rail impact test IAW MIL-STD-810. The Tool Outfit shall be positioned lengthwise, centered on a longitudinal centerline of a railway car. This railroad car carrying the Tool Outfit shall be impacted once each into the buffer cars at speeds of 7, 10, and 13 km/h with the Tool Outfit oriented in the forward direction and once at 13 km/h with the Tool Outfit oriented in the reverse direction. The rail impact procedure to be followed is described in MIL-STD-810, method no. 516.3, paragraphs I-3.10 and II-3.8 Procedure VIII - Rail Impact and including paragraph II-2.2. Examine the Tool Outfit and record any physical damage after each impact.

4.6.2.3 Functional test. The preproduction and initial production items shall be subjected to a functional test of set missions. A mission is defined as 2 hours of towing and three work tasks. Missions shall be performed in the following manner. Every 2 hours during towing in accordance with 4.6.2.1, stop and locate the tool outfit at two work sites. Using the instructions furnished with the tool outfit, set up for operation and perform the following three work tasks. Perform the above mission 20 times.

(1) First task.
  (a) Cut and deburr three tubes.
  (b) Flare tubing.
  (c) Bend tubing 90 degrees.
  (d) Wash the tube assemblies.
  (e) Proof pressure test the assemblies to 35000 kPa.

(2) Second task.
   (a) Cut and skive three hoses.
   (b) Assemble couplings to the hoses.
   (c) Wash the hose assemblies with solvents.
   (d) Proof pressure test the hose assemblies to 35000 kPa.

(3) Third task.
   (a) Transfer 190 Liter (L) of oil from one drum to another three times.
   (b) Prepare the tool outfit for movement to another location and continue the towing test.

Damage to or displacement (see 6.6.1) of any component, accessory, part or tool installed in or on the tool outfit, binding of any door, sliding table or landing leg, pinching or cutting of an electrical cable or hose, or evidence of defective welded, bolted or riveted connections, or any permanent deformation or damage or any malfunction a qualified operator or mechanic cannot remedy in 30 minutes with available tools and repair parts, shall constitute failure of this test.

4.6.2.4 Environmental.

4.6.2.4.1 Temperature test. Place the tool outfit in a chamber and connect to a 110-120 volt 50-60 Hz power source. Lower the chamber temperature to 255 K plus or minus 3 K, and maintain for 6 hours. Oil conforming to MIL-L-46167 may be used for low temperature tests. Perform the operations a. through k. below. Lower the chamber temperature to 230 K plus or minus 3 K and maintain for 6 hours. Raise the chamber temperature to 305 K, and maintain for 4 hours, then perform operations a. through k. below. Oil conforming to MIL-L-2104 shall be used for normal and high temperature tests. Raise the chamber temperature to 340 K plus or minus 3 K and maintain this temperature for 4 hours. Lower the chamber temperature to 302 K plus or minus 3 K and maintain this temperature for 4 hours, then perform operations a. through k. below:

a. Cut and skive hose.
b. Attach hose couplings.
c. Wash hose with solvent.
d. Proof pressure test hose to 35000 kPa.
e. Cut and deburr tubing.
f. Flare tubing.
g. Bend tubing.
h. Wash tubing with solvent.
i. Proof pressure test tubing to 35000 kPa.
j. Transfer 190 L of oil from one drum to another using the transfer pumps (MIL-L-46167 or equal).
k. Attach the multigage to the proof pressure tester and apply 24500 kPa.
l. Oil conforming to MIL-L-46167 may be used for low temperature test.

Failure of any tool to operate shall constitute failure of this test.

4.6.2.4.2 Rain. The tool outfit shall be parked on a horizontal surface with all doors closed and locked and shall be exposed to simulated rain falling on the tool outfit at an angle of approximately 45 degrees to the horizontal. The volume of rain impinging on the horizontal surface shall not be less than 1 L per minute per 1000 square cm of wetted horizontal area. Simulated rain shall originate from a source which

produces droplets of uniform distribution and velocity similar to natural rain at the point of contact with the unit. Expose the top and left side, top and front, top and right side, ant top and rear of the tool outfit for 5 minutes (total 20 minutes). Evidence of leakage of water into the interior or malfunction of an electrical component when attached to 110-120 volt, 50-60 Hz power upon completion of 20 minutes of rainfall shall constitute failure of this test. Note: The tester is cautioned to protect all personnel in the event of electrical malfunction.

4.6.2.5 Electrical interface.

4.6.2.5.1 Commercial. Connect the tool outfit power cord to 110-120 volt 50-60 Hz power drawn from a suitable power source. Operate each electrically driven tool and measure the voltage between the tool outfit body and ground. Failure of a tool to operate or voltage exceeding 5 volts between the body and ground shall constitute failure of this test.

4.6.2.5.2 Electrical interface, military. Repeat the test specified in 4.6.2.5.1 using the Military Standard Generator specified in 3.22.

4.6.2.6 Proof pressure test. Plug the proof pressure tester outlet and apply pressure of 70,000 kPa plus or minus 3500 kPa by pumping with the hand pump. Evidence of leakage or rupture shall constitute failure of this test.

4.6.2.7 Lifting and tiedown Test. To determine compliance with 3.11.1, the fully loaded Tool Outfit shall be subjected to the following tests.

4.6.2.7.1 Slinging Provisions. Each lifting provision including the connection to the structural members shall be tested. A static pull to no less than the required design load shall be conducted on each provision for no less than 90 seconds in a direction of pull 45 degrees upward form the Tool Outfit toward the apex where the four lifting cables meet in flight. All provisions do not have to be tested at the same time. Tests shall be performed IAW MIL-STD-209, including paragraph 5.5.2. The design loads for the four lifting provisions are as follows:

a. Front two provisions - 1,550 kilograms (kg).
b. Rear two provisions - 1,500 kg.

No permanent deformation or set or structural damage in the provisions or other equipment structural components shall result from application of the loads to the provisions.

4.6.2.7.2 Tie-down Provision. Each tie-down provision, including the connection to the structural members, shall be tested. A static pull to no less than the required design load shall be conducted on each provision for no less than six seconds in the direction of pull indicated. All provisions do not have to be tested at the same time. Test shall be performed IAW MIL-STD-209 including paragraph 5.5.3. The required design loads and directions of force for the four tie-down provisions are as follows:

a. All four provision - 1,360 kg each forward and rear longitudinal (horizontal toward the rear or forward, parallel to edge of bed).
b. All four provision - 680 kg each vertically downward.
c. All four provisions - 1,020 kg each lateral (horizontally outward, perpendicular force in a).

No permanent deformation or set or structural damage in the provisions or other equipment structural components shall result from application of the load to the provisions.

4.6.2.8 Noise level measurement. Measure the noise at the operator's station in accordance with MIL-STD-1474 for steady state noise while operating the hose cutting saw and the hose coupling machine while mounted in the tool outfit and with the tool outfit on a paved surface. Nonconformance to 3.18 shall constitute failure of this test.

4.6.2.9 Production test. Apply 70,000 kPa plus/minus 3,500 kPa to the proof pressure system with the hand pump. Open and close each door. Unpin and lower the wash tray. Raise and pin the wash tray in the stowed position. Unpin, lower, pin in place, unpin, raise and pin in place the support legs. Raise, pin in place, unpin and lower the rear upper door and the right hand side upper door. Wash the front, each side and rear of each tool outfit with a stream of water flowing at a minimum of four liters per minute. The water shall be concentrated on the riveted seams and edges of the doors. Time of wash shall be 10 minutes for each surface (rear, each side and front). Nonconformance to 3.8.3.9 or 3.10 or water present in the interior of the tool outfit shall constitute failure of this test.

4.7 Inspection comparison. The Government may select tool outfits at any time during the contract production period and subject these tool outfits to the examination specified in 4.6.1 and all tests specified in 4.6.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production tool outfits. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison tool outfit(s) to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison tool outfit(s) and shall be cause for refusal by the Government to continue acceptance of production tool outfits until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on tool outfits previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all tool outfits accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Preproduction pack inspection. The preproduction pack shall be examined for the defects specified in 4.8.2.2. Presence of one or more defects shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.2.2 Examination. The tool outfit shall be examined for the following defects.

110. Materials and methods not as specified for level A.
111. Equipment not placed and secured as provided for level A or Commercial.
112. Unprotected surfaces not preserved as specified for level A.
113. Trailer chassis components not preserved in accordance with the referenced document for level A.
114. Hand tools not preserved in accordance with referenced document for level A.
115. Loose hose and power cable not coiled and secured as specified for level A.
116. Consolidation not as specified for level A.
117. Doors and keys not secured as specified for level A.
118. Preservation not in accordance with referenced document for Commercial.
119. Packing not as specified.
120. Making missing, illegible, incorrect, or incomplete for level A, B, or Commercial.

5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination within the time frame specified (see 6.2.1) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.7). The preproduction pack may be accomplished utilizing either the preproduction model tool outfit or a production tool outfit. If the preproduction model tool outfit is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the government to facilitate comparison between the preproduction model tool outfit and production tool outfit.

5.2 Preservation. Preservation shall be level A or C as specified (see 6.2.1).

5.2.1 Level A.

5.2.1.1 Preservatives. Preservatives specified herein shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.2.1.2 Preservation methods. Methods of preservation specified herein shall conform to the applicable methods listed in MIL-P-116.

5.2.1.3 Sets or kits. When items are furnished in sets or kits with a carrying case, the items shall be preserved as specified herein and when possible replaced in the case.

5.2.1.4 Items having mounting brackets, holders, or tool boxes. Items furnished with mounting brackets, holders or contained in tool boxes shall be preserved as specified herein and when possible, replaced in the bracket, holder or tool box intended and secured to prevent movement or damage.

5.2.1.5 Unprotected surfaces. All unpainted, unfinished exterior ferrous metal surfaces of components, not specifically specified herein, requiring the application of a contact preservation in accordance with MIL-P-116 shall be preserved as follows:

5.2.1.5.1 Unfinished surfaces. Unfinished surfaces shall be coated with type P-1 or P-10 preservative.

5.2.1.5.2 Machined surfaces. Exposed exterior machined surfaces shall be coated with type P-11 preservative and the coated surface wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2 and secured in place with tape conforming to MIL-T-22085, type II.

5.2.1.6 Trailer chassis. The trailer chassis components shall be preserved in accordance with level A (mobile) requirements of MIL-V-62038.

5.2.1.7 Hand tools. All hand tools and similar equipment shall be preserved in accordance with the level A requirements of PPP-P-40.

5.2.1.8 Loose hose and power cable. Each loose 3/4 inch and smaller I.D. section of hose and the power cable shall be individually coiled and the coils secured with not less than three equally spaced ties of cotton twine or tape, then placed in the location provided. Hoses or cables shall not have any kinks or distortions and shall be placed in the location provided.

5.2.1.9 Preservation method IC-1 or IC-3. Items such as glove, filter elements, tape measures, safety glasses, electrical brushes, timer, tachometer, and impact wrench shall be preserved method IC-1 or IC-3. Items that are contained in suppliers' containers shall not be removed from the container.

5.2.1.10 Cans, jars and tubes. Items such as sealant, lubricant and oils contained in commercial cans, jars, or tubes shall not require additional preservation. Any glass containers shall be cushioned to prevent damage.

5.2.1.11 Miscellaneous supplies and equipment. Items, equipment, and components not specifically identified herein shall be preserved as specified for supplies and equipment of similar design and construction.

5.2.1.12 Consolidation. Components that cannot be placed in the tool boxes or secured by the mounting brackets or holders provided shall be consolidated together in close fitting boxes conforming to PPP-B-636, class weather-resistant. The boxes shall be of a size to fit into the available space within the body. The content shall be blocked, braced and cushioned as applicable to prevent movement and damage within the box. The box shall be secured within the body in a manner to prevent movement.

5.2.1.13 Keys and doors. Keys shall be preserved method IC-1 or IC-3 and secured in an obvious location when the rear door is opened. After completion of the preservation requirements all side doors shall be locked. The rear door shall remain unlocked.

5.2.2 Level C. Preservation of the trailer and all supplies and equipment shall be in accordance with MIL-STD-1190. Components shall be placed in the tool boxes or secured by the mounting brackets or holders provided to the greatest extent possible.

5.3 Packing (see 6.10). The trailer mounted tool outfit shall be prepared for shipment mobile in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules. Specifically, the rear door shall be secured by applying a band of strapping conforming to QQ-S-781, class 1, type I or IV, size 3/4 inch, finish B, horizontally and completely around the body, in a manner to secure the top section of the rear door from being opened. Fiberboard edge protectors shall be inserted under the strapping to prevent damage.

5.4 Marking. Marking shall be in accordance with MIL-STD-129. Bar code CCCCCCC marking are required and shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tool outfit is intended for use by personnel engaged in CCCCCCCCCCCC operations requiring the use of hydraulic test and repair tools for hydraulic systems of land used equipment.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Time frame required for submission of preproduction model (see 3.2).
c. When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
d. When initial production inspection is required and number of tool outfits to be furnished (see 3.3).
e. Time frame required for submission of the preproduction pack (see 5.1).
f. Degree of preservation required (see 5.2).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operations and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.3 Preproduction model. Any changes or deviations of production tool outfits from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to preserve, pack, and mark the tool outfits in accordance with this specification.

6.4 Government-furnished property. The contracting officer should arrange to furnish the contractor the M116A1 trailers in the quantities required in 3.21.

6.5 Government-loaned property. The contracting officer should arrange to loan the contractor the property specified in 3.22.

6.6 Definitions.

6.7 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the tool outfit in accordance with this specification.

6.8 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts list, wiring diagrams and contractor's maintenance and operation manual to be furnished with each tool outfit.

6.9 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of tool outfits. A suggested paragraph is as follows:

"Shipment of tool outfits shall include repair parts, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract."

6.10 Degrees of packings. Degrees of packing have not been included as the nature of the item is such that only mobile packing is appropriate. For purposes of preservation and packing level marking, the level of packing shall be considered level A.

6.11 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

6.12 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.13 Subject term (key word) listing.

AMSC number
Design
First article
Reclaimed materials

Custodian:
Army - AL

Preparing activity:
Army - AL

Review activities:
Army - ME
DLA - CS

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